# Ryan Ye

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## Education

**Cornell University** Ithaca, NY

*B.S. in Computer Science, College of Engineering, GPA 4.02* *Expected May 2028*

- **Coursework:** Deep Learning, Computer Vision, Machine Learning, Computer System Organization (C), Algorithms, Data Structures, Networks, Functional Programming (OCaml), Discrete Math, Linear Algebra
- **Honors:** BURE Fellow, CIDA Grant Recipient, AIMI Junior Affiliate, 4-time AIME Qualifier, USACO Silver Division

## Technical Skills

- **Languages:** Python, Java, C/C++, TypeScript/JavaScript, OCaml
- **ML:** PyTorch, Hugging Face Transformers, PEFT, scikit-learn, NumPy, pandas, OpenCV, YOLO
- **Tools:** Git, GitHub Actions, SLURM/HPC, Playwright, React, Node.js, Claude API

## Research Experience

**Undergraduate Researcher, Sun Lab at Cornell University** May 2025 – Present

*PI: Jennifer Sun. AI for science: scientific discovery agents and computer vision for animal behavior*

***Statistical validation for automated hypothesis discovery (AIMI Junior Affiliate)*** *Jun 2026 – Present*

- Implemented the randomization tests (p- and e-values for 11 claim types) and the false discovery rate reporting rules behind an independent check on what AI discovery agents report
- Ran five agent systems, 500+ runs in 13 test settings with known ground truth: the agents' own reports were 18 to 66% false; the check brought that to at most 1.3% for every agent; stress tests showed it can fail if held-out results leak

***Multi-lab video behavior recognition (MABe)*** *Feb 2026 – Present*

- Beat the frozen-backbone baseline on 9 of 15 lab datasets (13 of 15 on at least one split) by LoRA fine-tuning a video foundation model with a self-authored kinematic auxiliary loss; on one lab, video alone beat the pose-tracking ceiling
- Wrote and ran the 8-stage SLURM chain that took all 15 labs from raw video to evaluated 5-fold ensembles, staging up to 1.9 TB of features per lab on node-local NVMe; ran on Blackwell GPUs
- Switched the VideoPRISM backbone from JAX to a third-party PyTorch port and checked parity, which enabled LoRA fine-tuning; wrote the PyTorch training stage, video loader, and 21 unit tests

***Calf behavior monitoring (BURE Fellow, CIDA grant)*** *May 2025 – Present*

- Built a two-stage posture pipeline (YOLO11n detection, frozen DINO features) on 3,776 self-annotated images from 14 pens and 6 cameras: F1 0.93 with 300 labels, 0.98 with 2,988; 0.93+ on unseen cameras, calves, and days
- Rebuilt the evaluation (time-blocked test set, McNemar tests with Holm correction, block bootstrap), overturning two earlier claims; self-supervised features beat supervised ImageNet ones by 0.11–0.12 F1 at 50–100 labels
- Benchmarked five vision-language models (Qwen, Gemma 3, Gemini 2.5 Pro, PaliGemma, NVLM) for ear-tag and pose recognition; built a Jupyter workflow for veterinary researchers; first-author manuscript in preparation

## Publications

- M. Marks, …, **R. M. Ye**, et al. "A community benchmark enables high-resolution, cross-lab quantification of mouse social behavior." Under review at *Science*, 2026
- **R. M. Ye**, M. Madureira Ferreira, J. J. Sun, F. A. Leal Yepes. "Automating pose detection in calves: a computer vision framework for non-invasive behavioral monitoring." Poster, ADSA 2026
- M. Madureira Ferreira, …, **R. M. Ye**, J. J. Sun, F. A. Leal Yepes. "A smarter lung ultrasonography solution for lung consolidation classification in dairy calves using a deep learning approach." *J. Dairy Sci.*, 2026

## Projects

**DoRA Re-Implementation and Evaluation** Spring 2026

*CS 4782 Deep Learning final project, 4-person team. PyTorch, Hugging Face PEFT* *GitHub*

- Team re-implementation of DoRA, tested against LoRA on Qwen2.5-3B (AMC12, MATH); ran all HPC experiments
- Built the evaluation harness (logit-probe and two-pass scoring, separating what a model knows from what it generates), the data pipeline with a leakage filter, and the figures; diagnosed a critical bug in the DoRA layer

**Worship Rig** Sep 2026

*Personal project. Electron, JavaScript, Web Audio* *Site · GitHub*

- Built and shipped a Mac app that turns a MIDI keyboard into a keys rig for worship bands: 43 instruments, a key drone, setlists, per-sound EQ, and WAV recording, on a Web Audio engine that also renders offline for tests
- Wrote 1,000+ automated tests (Playwright), a GarageBand sample converter, and CI builds for Apple Silicon and Intel

**Sisyphus: AI Productivity App (Claude Builders Club Hackathon, 4-person team)** Nov 2025

- Checks task photos with Claude; one of 5 winners among 38 projects (Fun Award); worked on frontend and AI validation

## Leadership

**Basic Coding**, Co-Founder & President. Free coding classes for 300+ students in 8 states Aug 2020 – Jun 2024